

TP5 – Vaultwarden via Docker

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Pour l'installation de Docker Engine, j'utilise la méthode via apt donné indiqué par le site officiel.

```
root@Debian:~# sudo apt update
sudo apt install ca-certificates curl
```

```
root@Debian:~# sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/debian/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
```

```
root@Debian:~# sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/debian/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
root@Debian:~# sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/debian
Suites: $(. /etc/os-release && echo "$VERSION_CODENAME")
Components: stable
Signed-By: /etc/apt/keyrings/docker.asc
EOF
Types: deb
URIs: https://download.docker.com/linux/debian
Suites: trixie
Components: stable
Signed-By: /etc/apt/keyrings/docker.asc
root@Debian:~#
```

```
root@Debian:~# apt update
Atteint : 1 http://deb.debian.org/debian trixie InRelease
Atteint : 2 http://deb.debian.org/debian trixie-updates InRelease
Atteint : 3 http://security.debian.org/debian-security trixie-security InRelease
Atteint : 4 https://download.docker.com/linux/debian trixie InRelease
Tous les paquets sont à jour.
root@Debian:~#
```

```
root@Debian:~# sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Le paquet suivant a été installé automatiquement et n'est plus nécessaire :
  linux-image-6.12.38+deb13-amd64
Veuillez utiliser « sudo apt autoremove » pour le supprimer.

Installation de :
  containerd.io  docker-buildx-plugin  docker-ce  docker-ce-cli  docker-compose-plugin

Installation de dépendances :
  docker-ce-rootless-extras  iptables  libip4tc2  libnftnl0  libslirp0  slirp4netns
  git  libcurl3t64-gnutls  libip6tc2  libngtcp2-16  patch
  git-man  liberror-perl  libnetfilter-conntrack3  libngtcp2-crypto-gnutls8  pigz

Paquets suggérés :
  cgroupfs-mount  docker-model-plugin  git-email  gitk  git-cvs  git-svn  ed
  | cgroup-lite  git-doc  git-gui  gitweb  git-mediawiki  firewallld  diffutils-doc

Sommaire :
  Mise à niveau de : 0. Installation de : 21Supprimé : 0. Non mis à jour : 0
Taille du téléchargement : 103 MB
Espace nécessaire : 420 MB / 28,7 GB disponible
```

```
root@Debian:~# sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Mon 2025-12-08 16:48:18 CET; 25s ago
 Invocation: 2c3ab4cd546040f19135afaacc8e1090
 TriggeredBy: ● docker.socket
    Docs: https://docs.docker.com
   Main PID: 9828 (dockerd)
     Tasks: 9
    Memory: 26.1M (peak: 27.5M)
       CPU: 1.483s
    CGroup: /system.slice/docker.service
            └─9828 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

déc. 08 16:48:15 Debian dockerd[9828]: time="2025-12-08T16:48:15.906092848+01:00" level=info m
déc. 08 16:48:16 Debian dockerd[9828]: time="2025-12-08T16:48:16.118695104+01:00" level=info m
déc. 08 16:48:16 Debian dockerd[9828]: time="2025-12-08T16:48:16.138514744+01:00" level=info m
déc. 08 16:48:18 Debian dockerd[9828]: time="2025-12-08T16:48:18.044595694+01:00" level=info m
déc. 08 16:48:18 Debian dockerd[9828]: time="2025-12-08T16:48:18.073220009+01:00" level=info m
déc. 08 16:48:18 Debian dockerd[9828]: time="2025-12-08T16:48:18.073678574+01:00" level=info m
déc. 08 16:48:18 Debian dockerd[9828]: time="2025-12-08T16:48:18.180469688+01:00" level=info m
déc. 08 16:48:18 Debian dockerd[9828]: time="2025-12-08T16:48:18.232092409+01:00" level=info m
déc. 08 16:48:18 Debian dockerd[9828]: time="2025-12-08T16:48:18.232384707+01:00" level=info m
déc. 08 16:48:18 Debian systemd[1]: Started docker.service - Docker Application Container Engi
```

```
root@Debian:~# sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
ea52d2000f90: Download complete
Digest: sha256:f7931603f70e13dbd844253370742c4fc4202d290c80442b2e68706d8f33ce26
Status: Downloaded newer image for hello-world:latest
```

```
Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.
```

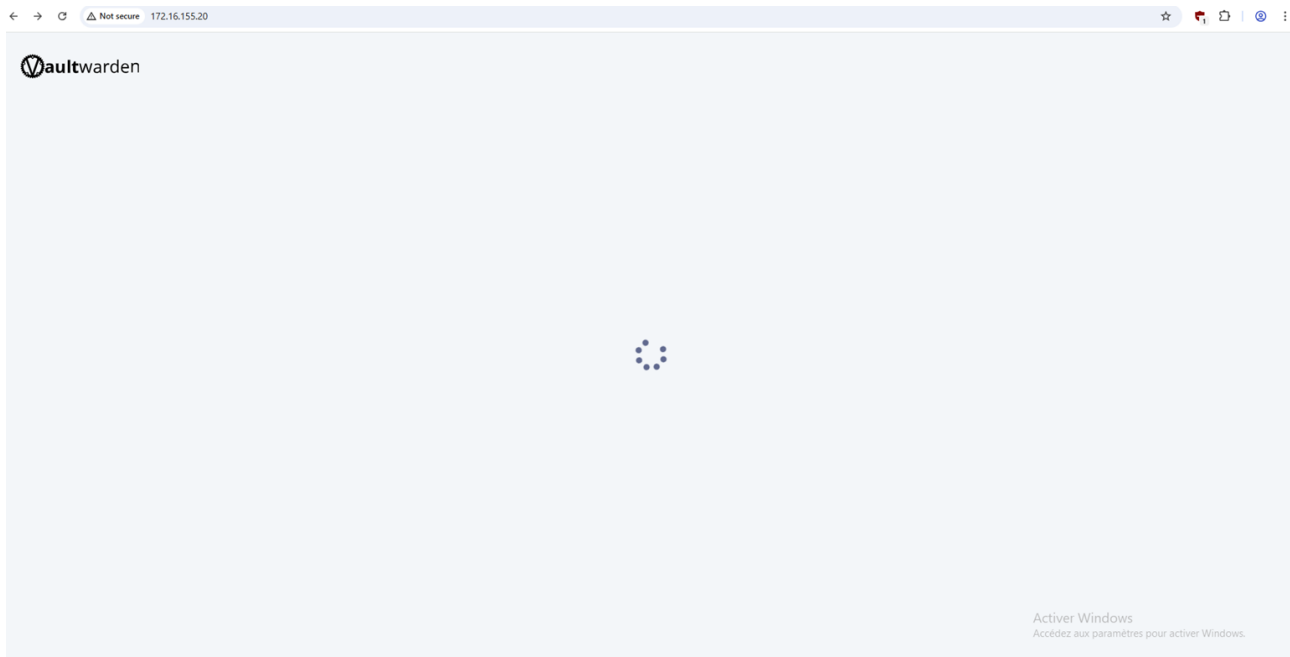
```
To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash
```

```
Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/
```

```
For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

```
root@Debian:~# docker pull vaultwarden/server:latest
latest: Pulling from vaultwarden/server
aa7aba5b8287: Pull complete
59e22667830b: Pull complete
cc18e03c8cd8: Pull complete
f6e0865da50e: Pull complete
6bec2bc6dc05: Pull complete
f2bd28c8128d: Download complete
Digest: sha256:84fd8a47f58d79alad824c27be0a9492750c0fa5216b35c749863093bfa3c3d7
Status: Downloaded newer image for vaultwarden/server:latest
docker.io/vaultwarden/server:latest
```

```
root@Debian:~# docker run -d --name vaultwarden -v /vw-data://data/ -p 80:80 vaultwarden/server:latest
e5ca54d3021e728095576e39344971605a53df70c1dc2f2301dac2646efe4e50
```



Malheureusement, VaultWarden ne fonctionne pas car les navigateurs bloquent une ressource utilisée par le logiciel. Un moyen de contourner cela est en passant par HTTPS, pour cela je vais suivre la méthode proposée par VaultWarden en installant mkcert via homebrew

```
root@Debian:~# sudo apt install libnss3-tools
Le paquet suivant a été installé automatiquement et n'est plus nécessaire :
  linux-image-6.12.38+debl3-amd64
Veuillez utiliser « sudo apt autoremove » pour le supprimer.

Installation de :
  libnss3-tools

Installation de dépendances :
  libnspr4 libnss3

Sommaire :
  Mise à niveau de : 0. Installation de : 3 Supprimé : 0. Non mis à jour : 0
Taille du téléchargement : 20589 kB
Espace nécessaire : 10,1 MB / 27,9 GB disponible
```

```

root@Debian:~# sudo apt install build-essential procps curl file git
procps est déjà la version la plus récente (2:4.0.4-9).
curl est déjà la version la plus récente (8.14.1-2+deb13u2).
file est déjà la version la plus récente (1:5.46-5).
git est déjà la version la plus récente (1:2.47.3-0+deb13u1).
git passé en « installé manuellement ».
Le paquet suivant a été installé automatiquement et n'est plus nécessaire :
  linux-image-6.12.38+deb13-amd64
Veuillez utiliser « sudo apt autoremove » pour le supprimer.

Installation de :
  build-essential

Installation de dépendances :
  binutils                g++-14                libasan8               libfile-fcntllock-perl  libquadmth0
  binutils-common         g++-14-x86-64-linux-gnu  libbinutils            libgcc-14-dev           libstdc++-14-dev
  binutils-x86-64-linux-gnu  g++-x86-64-linux-gnu  libc-dev-bin          libgomp1                libstdc++-14-dev
  cpp                     gcc                   libc6-dev             libgprofng0             libtsan2
  cpp-14                  gcc-14                libc6-dev             libhwasan0              libubsan1
  cpp-14-x86-64-linux-gnu  gcc-14-x86-64-linux-gnu  libcrypt-dev          libisl23                 linux-libc-dev
  cpp-x86-64-linux-gnu     gcc-x86-64-linux-gnu  libctf-nobfd0         libitm1                  make
  dpkg-dev                 libalgorithm-diff-perl  libctf0               liblsan0                 manpages-dev
  fakeroot                 libalgorithm-diff-xs-perl  libdpkg-perl          libmpc3                  rpcsvc-proto
  g++                      libalgorithm-merge-perl  libfakeroot           libmpfr6

Paquets suggérés :
  binutils-doc    gcc-14-locales      g++-multilib      autoconf    bison          gdb-x86-64-linux-gnu  libstdc++-14-doc
  gprofng-gui     cpp-14-doc          g++-14-multilib  automake    gdb            libc-devtools        make-doc
  binutils-gold   debian-keyring      gcc-14-doc        libtool     gcc-doc        glibc-doc
  cpp-doc         debian-tag2upload-keyring  gcc-multilib      flex        gcc-14-multilib  bzip2

Sommaire :
  Mise à niveau de : 0. Installation de : 50Supprimé : 0. Non mis à jour : 0
Taille du téléchargement : 76,1 MB
Espace nécessaire : 287 MB / 27,9 GB disponible

```

Pour installer Homebrew il ne faut pas être sur root ni utiliser sudo

```

neo@Debian:~$ /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
==> Checking for `sudo` access (which may request your password)...
==> This script will install:
/home/linuxbrew/.linuxbrew/bin/brew
/home/linuxbrew/.linuxbrew/share/doc/homebrew
/home/linuxbrew/.linuxbrew/share/man/man1/brew.1
/home/linuxbrew/.linuxbrew/share/zsh/site-functions/_brew
/home/linuxbrew/.linuxbrew/etc/bash_completion.d/brew
/home/linuxbrew/.linuxbrew/Homebrew
==> The following new directories will be created:
/home/linuxbrew/.linuxbrew/bin
/home/linuxbrew/.linuxbrew/etc
/home/linuxbrew/.linuxbrew/include
/home/linuxbrew/.linuxbrew/lib
/home/linuxbrew/.linuxbrew/sbin
/home/linuxbrew/.linuxbrew/share
/home/linuxbrew/.linuxbrew/var
/home/linuxbrew/.linuxbrew/opt
/home/linuxbrew/.linuxbrew/share/zsh
/home/linuxbrew/.linuxbrew/share/zsh/site-functions
/home/linuxbrew/.linuxbrew/var/homebrew
/home/linuxbrew/.linuxbrew/var/homebrew/linked
/home/linuxbrew/.linuxbrew/Cellar
/home/linuxbrew/.linuxbrew/Caskroom
/home/linuxbrew/.linuxbrew/Frameworks

Press RETURN/ENTER to continue or any other key to abort:

```

```

neo@Debian:~$ echo >> /home/neo/.bashrc
echo 'eval "$(/home/linuxbrew/.linuxbrew/bin/brew shellenv)'" >> /home/neo/.bashrc
eval "$(/home/linuxbrew/.linuxbrew/bin/brew shellenv)"
neo@Debian:~$ brew install gcc
=> Fetching downloads for: gcc
✓ Bottle Manifest gcc (15.2.0) [Downloaded 23.4KB/ 23.4KB]
✓ Bottle Manifest gmp (6.3.0) [Downloaded 13.3KB/ 13.3KB]
✓ Bottle gmp (6.3.0) [Downloaded 1.2MB/ 1.2MB]
✓ Bottle Manifest isl (0.27) [Downloaded 14.5KB/ 14.5KB]
✓ Bottle isl (0.27) [Downloaded 2.3MB/ 2.3MB]
✓ Bottle Manifest mpfr (4.2.2) [Downloaded 12.7KB/ 12.7KB]
✓ Bottle mpfr (4.2.2) [Downloaded 1.3MB/ 1.3MB]
✓ Bottle Manifest libmpc (1.3.1) [Downloaded 16.8KB/ 16.8KB]
✓ Bottle libmpc (1.3.1) [Downloaded 180.7KB/180.7KB]
✓ Bottle Manifest lz4 (1.10.0) [Downloaded 13.8KB/ 13.8KB]
✓ Bottle lz4 (1.10.0) [Downloaded 320.8KB/320.8KB]
✓ Bottle Manifest xz (5.8.1) [Downloaded 14.5KB/ 14.5KB]
✓ Bottle Manifest zlib (1.3.1) [Downloaded 10.7KB/ 10.7KB]
✓ Bottle xz (5.8.1) [Downloaded 816.7KB/816.7KB]
✓ Bottle Manifest zstd (1.5.7) [Downloaded 16.2KB/ 16.2KB]
✓ Bottle zlib (1.3.1) [Downloaded 175.1KB/175.1KB]
✓ Bottle zstd (1.5.7) [Downloaded 1.1MB/ 1.1MB]
✓ Bottle Manifest binutils (2.45.1) [Downloaded 41.0KB/ 41.0KB]
✓ Bottle binutils (2.45.1) [Downloaded 93.9MB/ 93.9MB]
✓ Bottle gcc (15.2.0) [Downloaded 162.0MB/162.0MB]
=> Installing dependencies for gcc: gmp, isl, mpfr, libmpc, lz4, xz, zlib, zstd and binutils
=> Installing gcc dependency: gmp
=> Pouring gmp--6.3.0.x86_64_linux.bottle.tar.gz
📦 /home/linuxbrew/.linuxbrew/Cellar/gmp/6.3.0: 24 files, 4.1MB
=> Installing gcc dependency: isl
=> Pouring isl--0.27.x86_64_linux.bottle.tar.gz
📦 /home/linuxbrew/.linuxbrew/Cellar/isl/0.27: 75 files, 10.3MB
=> Installing gcc dependency: mpfr
=> Pouring mpfr--4.2.2.x86_64_linux.bottle.tar.gz
📦 /home/linuxbrew/.linuxbrew/Cellar/mpfr/4.2.2: 32 files, 4.2MB
=> Installing gcc dependency: libmpc
=> Pouring libmpc--1.3.1.x86_64_linux.bottle.tar.gz
📦 /home/linuxbrew/.linuxbrew/Cellar/libmpc/1.3.1: 14 files, 658.4KB
=> Installing gcc dependency: lz4
=> Pouring lz4--1.10.0.x86_64_linux.bottle.1.tar.gz
📦 /home/linuxbrew/.linuxbrew/Cellar/lz4/1.10.0: 24 files, 817.8KB
=> Installing gcc dependency: xz
=> Pouring xz--5.8.1.x86_64_linux.bottle.tar.gz
📦 /home/linuxbrew/.linuxbrew/Cellar/xz/5.8.1: 97 files, 2.7MB
=> Installing gcc dependency: zlib
=> Pouring zlib--1.3.1.x86_64_linux.bottle.tar.gz
📦 /home/linuxbrew/.linuxbrew/Cellar/zlib/1.3.1: 14 files, 487.4KB
=> Installing gcc dependency: zstd
=> Pouring zstd--1.5.7.x86_64_linux.bottle.1.tar.gz
📦 /home/linuxbrew/.linuxbrew/Cellar/zstd/1.5.7: 32 files, 3.2MB
=> Installing gcc dependency: binutils
=> Pouring binutils--2.45.1.x86_64_linux.bottle.tar.gz

```

```

neo@Debian:~$ brew install mkcert
=> Fetching downloads for: mkcert
✓ Bottle Manifest mkcert (1.4.4) [Downloaded 13.2KB/ 13.2KB]
✓ Bottle mkcert (1.4.4) [Downloaded 1.5MB/ 1.5MB]
=> Pouring mkcert--1.4.4.x86_64_linux.bottle.tar.gz
📦 /home/linuxbrew/.linuxbrew/Cellar/mkcert/1.4.4: 7 files, 3.4MB
=> Running 'brew cleanup mkcert'...
Disable this behaviour by setting 'HOMEBREW_NO_INSTALL_CLEANUP=1'.
Hide these hints with 'HOMEBREW_NO_ENV_HINTS=1' (see 'man brew').

```

```

neo@Debian:~$ mkcert -install
Created a new local CA ✨
Sudo password:
The local CA is now installed in the system trust store! ⚡

```

```
neo@Debian:~$ mkcert 155.sio "*.155.sio" 155.sio localhost 127.0.0.1 ::1

Created a new certificate valid for the following names
- "155.sio"
- "*.155.sio"
- "155.sio"
- "localhost"
- "127.0.0.1"
- "::1"

Reminder: X.509 wildcards only go one level deep, so this won't match a.b.155.sio

The certificate is at "./155.sio+5.pem" and the key at "./155.sio+5-key.pem"

It will expire on 8 March 2028
```

```
root@Debian:~# docker stop vaultwarden
vaultwarden
root@Debian:~# docker container prune
WARNING! This will remove all stopped containers.
Are you sure you want to continue? [y/N] y
Deleted Containers:
037a3e53eb137bel81de680d14ad84be423fe66a2dl38c4e70a86c9e2bc9b81f

Total reclaimed space: 8.192kB
root@Debian:~#
```